Brian Saastad

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SENIOR SOLUTIONS ARCHITECT

Turning ambiguous requirements into production-ready architecture

Senior Solutions Architect with extensive experience designing and building cloud and hybrid enterprise platform architectures for business-critical production environments across fintech and customer communications. Known for taking ownership of complex technical problems without established internal patterns, shaping practical cloud solutions around business and cost constraints, and carrying secure, scalable systems from concept into production.

Career Highlights

- **Delivered architecture for complex technical problems for greenfield and first-time technical initiatives**, including building out looch's immature AWS backend and advancing an on-prem Universal Repository into a hybrid AWS/on-premises enterprise system.
- **Converted fragmented systems and custom workflows into reusable enterprise platforms that reduced dependency on bespoke builds**, such as Universal Repository for 200+ systems and File Manager's reusable client implementation framework.
- **Reduced production risk across business-critical systems by improving availability, recovery, and operational visibility**, as shown in prevention of up to $150K in annual contractual fines and recovery time reduction by up to 99%.

KEY SKILLS >>> Solution Architecture | Cloud & Hybrid Architecture | Enterprise Platform Design | Requirements Translation
Stakeholder Alignment | Technical Decision-Making | Production Readiness | Security & Compliance Alignment | Technical Leadership

AWS Serverless Architecture | REST API Design | Microservices Architecture | Event-Driven Architecture | Infrastructure as Code
CI/CD | Cloud Security & IAM | VPC Connectivity | Observability & Log Monitoring | Data Storage Architecture | Hybrid Cloud / On-Prem

Professional Experience

looch, Las Vegas, NV | *Financial services startup*

SENIOR SOLUTIONS ARCHITECT — 2024 – Present

Tapped by co-founder to build out immature backend and cloud infrastructure of fintech startup's mobile platform and core services. Translated regulatory and integration requirements into secure, observable, deployment-ready AWS architectures designed for production scale and cost control. Served as hands-on architect for ambiguous technical problems, selecting tools and defining implementation approach while validating AI-assisted development output for production readiness.

- **Architected reusable FDX-compliant open banking platform on AWS Lambda and API Gateway** that gave Plaid and future financial data aggregators standardized access to authorized customer account data through secure OAuth-based flows, reducing future company-side aggregator onboarding from a 6- to 8-week first build to <2 hours with no code changes.
- **Designed and built a cloud-native IRS IRIS 1099 filing platform on Python AWS Lambda and AWS Batch**, converting complex tax-reporting requirements into a production-validated, $2-per-submission recurring revenue product with automated form determination, IRS-compliant submission, schema validation, and recipient PDF delivery.
- **Built centralized observability platform across all backend services, including AWS Lambda and batch workloads**, using fault-isolated dual-pipeline design that separated real-time error alerting from long-term log archival and enabled sub-minute error alerts and seven years of searchable log history through S3/Glue/Athena at low cost.
- **Hardened AWS VPC connectivity and cloud security posture** by redesigning IAM permissions and service-to-service access, including VPC endpoints, security groups, and routing rules to reduce attack surface, remove implicit trust, and enforce least-privilege private-network communication.
- **Reduced production recovery time from 35–40 minutes to 30 seconds across a 92-function Lambda deployment unit** by building an alias-based promotion and rollback platform that decoupled code deployments from production traffic shifts and unified CLI/UI actions under one audit trail.

Broadridge Customer Communications (Formally DST Output), El Dorado Hills, CA | *Customer communications company*

PRINCIPAL SOFTWARE ARCHITECT (PROFESSIONAL) — 2016 – 2023

Retained during acquisition to own development of Universal Repository (UR) - a strategic data platform supporting one of US' largest customer communications operations. Modernized platform from DST-era infrastructure foundation into hybrid AWS/on-premises enterprise system, in line with Broadridge's Target State Architecture.

- **Scaled Universal Repository into multi-tenant, active-active, highly available enterprise data platform** capable of serving as the central storage, retrieval, query, and management layer for Broadridge's **200+** customer communications systems.
- **Established single REST API contract for Universal Repository data ingest, search, and retrieval**, enabling consuming applications to utilize consistent integration model whether customer data resided in AWS or remained on premises for contractual, compliance, or data residency reasons.
- **Enabled Universal Repository to support scalable storage, metadata search, full-text search, low per-request cost, and active-active resilience across AWS and on-premises deployments** by separating core data functions into independently scalable layers and pairing cloud-native services with parallel data-center components.
- **Supported Universal Repository's five-nines availability target -** limiting expected unplanned downtime to **~5** minutes per year outside scheduled maintenance windows **-** by designing AWS deployment across **2** regions and **4+** availability zones.

DST Output, El Dorado Hills, CA | *Acquired by Broadridge*

SENIOR SOFTWARE ENGINEER — 2008 – 2016

Recruited by VP to join elite infrastructure engineering team delivering complex, multi-year systems for one of the country's largest customer communications operations. Contributed to architecture and development of shared infrastructure systems, including early Universal Repository work, supporting file movement, production monitoring, and job orchestration across high-volume print/mail workflows.

- **Strengthened reliability across independent print/mail production systems** by designing Sonar, a file-tracking platform that monitored expected versus actual client data movement, tracked downstream branches across multiple production paths, and surfaced stalled jobs before missed due-out dates, helping prevent up to **$150K** in annual contractual fines.
- **Built File Manager, an event-driven orchestration framework for inbound Connect:Direct and SFTP files** that replaced custom per-client pipeline builds with reusable, pre-validated modules for common ingestion, processing, and output workflows; reduced client implementation timelines from 2–4 months on average to as fast as 2–3 days for simple clients.
    - Improved production visibility within File Manager by building single-pane-of-glass monitoring interface that gave operators real-time status across active client jobs instead of requiring checks across multiple systems.
- **Prevented misaligned infrastructure builds by helping stakeholders distinguish stated requests from actual operational needs**, then translating clarified requirements into technically viable solutions.
    - **Avoided unnecessary non-rewritable storage implementation with long-term cost and operational implications for a client** by clarifying request for WORM storage and surfacing actual need for user-level read access controls.
- **Established shared API standards and reusable integration patterns** for development teams.

Visualize, Rancho Cordova, CA | Privately owned data visualization and graphing tools tech company

SENIOR SOFTWARE ENGINEER — 2006 - 2008

Joined to support expansion of data visualization and graphing product line across stock tracking, genealogy mapping, and statistical analysis use cases. Contributed across application development, release packaging, client-facing technical support, and technical stakeholder communication. Later assumed interim technical leadership responsibilities, bridging client and technical stakeholders.

- **Built and released Pedigree Prism, an end-user genealogy mapping application** that transformed raw inheritance-trait data into visual relationship models, including product packaging and publication through an e-commerce portal.
- **Developed and supported stock-market and statistical visualization tools**, enabling users to model data in graphing formats, drill into relationships and patterns, and resolve client-facing technical issues.

ADDITIONAL EXPERIENCE

Software Engineer | Output Technology Solutions (formally International Billing Systems)
Developer | Quality Medical Adjudication (acquired by Health Management Systems)

Certifications & Professional Development

AWS: Solutions Architect – Associate | Developer – Associate | SysOps Administrator – Associate
Data Engineer – Associate | AI Practitioner | Cloud Practitioner
DataStax: Apache Cassandra Professional Certification
U.C. Davis Extension: Relational Data Modeling

Technical Proficiencies

AWS Cloud: Lambda, API Gateway, S3, CloudWatch, Kinesis, EventBridge, SQS, SNS, CloudFront, Cognito, Glue, Athena, AWS Batch
AWS Security & Networking: IAM, VPC, VPC Endpoints
Databases & Search: DynamoDB, Oracle, Solr, Cassandra, Elasticsearch, MySQL
Programming Languages: Python, TypeScript, Node.js, Java, C, Perl
DevOps & Build Tools: AWS CloudFormation, Serverless Framework, Jenkins, Maven, CI/CD Pipelines, Kubernetes
APIs & Integration: REST APIs, OAuth 2.0, PKCE, FDX, Plaid, Connect:Direct, SFTP
AI Tools: Claude Code